

Koiter asymptotic analysis about a nonlinear pre-buckling state

Implementation notes for `NLprebuck=True` in `bfsccylinder_models`

Abstract

This note documents the implementation of the single-mode Koiter method with a nonlinear pre-buckling state in `koiter_cylinder_newton_raphson.py` (von Kármán kinematics) and `koiter_cylinder_newton_raphson_sanders.py` (Sanders kinematics). It covers the incremental-iterative algorithm that locates the expansion point in the neighbourhood of the bifurcation load, the consistency requirements on the operators used in the eigenvalue problem and in the asymptotic expansion, and the stiffness-weighted orthogonality condition imposed on the second-order fields. Equation numbers of the form “Sun Eq. (nn)” refer to Sun, Tian, Li and Wang, *Accelerated Koiter method for post-buckling analysis of thin-walled shells under axial compression*, Thin-Walled Structures **155** (2020) 106962.

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1 Scope

The routine `fkoiter_cyl_SS3` performs, for a circular cylindrical shell under axial compression with SS-3 boundary conditions,

1. a pre-buckling analysis,
2. a buckling eigenvalue analysis about the pre-buckling state,
3. the Koiter asymptotic expansion about the bifurcation point, returning the post-buckling coefficients a_{ijk} and b_{ijkl} .

With `NLprebuck=False` the pre-buckling state is the linear elastic solution and the pre-buckling path is exactly linear in the load parameter. With `NLprebuck=True` the pre-buckling state is obtained from a geometrically nonlinear analysis, and the whole expansion must then be carried out about a state *in the neighbourhood of the bifurcation point*. This document describes the latter case.

Throughout, λ is the load parameter multiplying the reference load N_{xx}^{unit} , $\mathbf{f}_0$ is the corresponding reference load vector, λ_b is the load level at which the pre-buckling state is evaluated, and λ_c is the bifurcation load.

2 Kinematics and the discrete energy functional

2.1 Strain measures

The membrane strains are written, for both kinematic models, as

$$\boldsymbol{\varepsilon} = \mathbf{B}_m \mathbf{q} + \frac{1}{2} \begin{Bmatrix} g_1^2 \\ g_2^2 \\ 2 g_1 g_2 \end{Bmatrix}, \quad \boldsymbol{\kappa} = \mathbf{B}_b \mathbf{q}, \quad (1)$$

where g_1 and g_2 are the two gradient measures that carry the geometric nonlinearity,

$$\text{von Kármán: } g_1 = w_{,x}, \quad g_2 = w_{,y}, \quad \text{Sanders: } g_1 = w_{,x}, \quad g_2 = w_{,y} - \frac{v}{R}. \quad (2)$$

Every nonlinear term of the Sanders membrane strains is recovered from the von Kármán one by the substitution $(w_{,x}, w_{,y}) \rightarrow (g_1, g_2)$, which is why the two modules differ only in the definition (2) and in the corresponding interpolation matrices $\mathbf{G}_1, \mathbf{G}_2$ with $g_\alpha = \mathbf{G}_\alpha \mathbf{q}$.

The constitutive relation is the usual laminated-shell one, $\mathbf{N} = \mathbf{A}\boldsymbol{\varepsilon} + \mathbf{B}\boldsymbol{\kappa}$ and $\mathbf{M} = \mathbf{B}\boldsymbol{\varepsilon} + \mathbf{D}\boldsymbol{\kappa}$, collectively denoted $\mathbf{H}$.

2.2 State and rates are independent fields

The pre-buckling state is a curve $\mathbf{q}_0(\lambda)$. The asymptotic expansion needs, besides the state itself, its first and second derivatives with respect to λ , written $\dot{\mathbf{q}}_0$ and $\ddot{\mathbf{q}}_0$. Differentiating (1) along the path, and abbreviating $g_\alpha^s = \mathbf{G}_\alpha \mathbf{q}_0$, $g_\alpha^d = \mathbf{G}_\alpha \dot{\mathbf{q}}_0$, $g_\alpha^{dd} = \mathbf{G}_\alpha \ddot{\mathbf{q}}_0$,

$$\dot{\boldsymbol{\varepsilon}} = \mathbf{B}_m \dot{\mathbf{q}}_0 + \begin{Bmatrix} g_1^s g_1^d \\ g_2^s g_2^d \\ g_1^s g_2^d + g_2^s g_1^d \end{Bmatrix}, \quad \dot{\boldsymbol{\kappa}} = \mathbf{B}_b \dot{\mathbf{q}}_0, \quad (3)$$

$$\ddot{\boldsymbol{\varepsilon}} = \mathbf{B}_m \ddot{\mathbf{q}}_0 + \begin{Bmatrix} (g_1^d)^2 + g_1^s g_1^{dd} \\ (g_2^d)^2 + g_2^s g_2^{dd} \\ 2g_1^d g_2^d + g_1^s g_2^{dd} + g_2^s g_1^{dd} \end{Bmatrix}, \quad \ddot{\boldsymbol{\kappa}} = \mathbf{B}_b \ddot{\mathbf{q}}_0, \quad (4)$$

and the derivatives with respect to the nodal degrees of freedom

$$\frac{\partial \varepsilon}{\partial \mathbf{q}_a} = \mathbf{B}_m + \left\{ \begin{array}{c} g_1^s \mathbf{G}_1 \\ g_2^s \mathbf{G}_2 \\ g_1^s \mathbf{G}_2 + g_2^s \mathbf{G}_1 \end{array} \right\}, \quad \frac{\partial^2 \varepsilon}{\partial \lambda \partial \mathbf{q}_a} = \left\{ \begin{array}{c} g_1^d \mathbf{G}_1 \\ g_2^d \mathbf{G}_2 \\ g_1^d \mathbf{G}_2 + g_2^d \mathbf{G}_1 \end{array} \right\}. \quad (5)$$

Remark. If one assumes the pre-buckling path to be linear in the load parameter, $\mathbf{q}_0(\lambda) = \lambda \mathbf{q}_0(1)$, then $\dot{\mathbf{q}}_0 = \mathbf{q}_0(1)$, $\ddot{\mathbf{q}}_0 = \mathbf{0}$ and $g_\alpha^s = \lambda g_\alpha^d$, so (3)–(5) collapse onto expressions containing λ explicitly. That assumption is precisely what removes the nonlinear pre-buckling behaviour from the expansion, and it is not made here: state, first rate and second rate are carried as three independent nodal fields.

2.3 Discrete energy derivatives

Let ϕ be the total potential energy. The following contractions are assembled element by element and are the building blocks of the expansion:

symbol in the code	meaning
phi2	$\phi_2 = \partial^2 \phi / \partial \mathbf{q}^2$, the tangent operator at the critical state
phi3_ab[(i,j)]	ϕ_3 contracted with modes i, j , one free index
phi4[(i,j,k,l)]	ϕ_4 fully contracted
phi20_a[i]	$\partial \phi_2 / \partial \lambda$ contracted with mode i , one free index
phi30_ab[(i,j)]	$\partial \phi_3 / \partial \lambda$ contracted with modes i, j , one free index
phi200_ab[(i,j)]	$\partial^2 \phi_2 / \partial \lambda^2$ fully contracted

All of them are built from (3)–(5) together with the constant quadratic form $\partial^2 \varepsilon / \partial \mathbf{q}_a \partial \mathbf{q}_b$, which for (1) is $\mathbf{G}_1^\top \mathbf{G}_1$, $\mathbf{G}_2^\top \mathbf{G}_2$ and $\mathbf{G}_1^\top \mathbf{G}_2 + \mathbf{G}_2^\top \mathbf{G}_1$ in the three strain components.

3 Locating the expansion point

3.1 Why the load level matters

Koiter’s expansion is asymptotic about the bifurcation point. Sun §2.2 requires the pre-buckling state to satisfy $\lambda_b / \lambda_c \approx 0.995$. If instead the pre-buckling problem is solved once at the reference load and the eigenvalue problem is posed about that state, then for the shells considered here λ_b / λ_c is of the order of 0.1. At that load level the radial pre-buckling deflection is one to three per cent of a wall thickness, the boundary layer produced by the edge restraint has not developed, and $\|\mathbf{K}_C^{NL}(\mathbf{q}_0)\| / \|\mathbf{K}_{C_0}\| \sim 2 \times 10^{-4}$; the eigenvalue analysis is then a linear buckling analysis in all but name and the whole computation returns the *membrane* pre-buckling answer.

3.2 Iterative eigenvalue algorithm

The expansion point is located by the incremental-iterative scheme of Sun Eqs. (44)–(46). At the current load level λ_b the pre-buckling state $\mathbf{q}_0$ is known and the eigenvalue problem

$$[\mathbf{K}_C(\mathbf{q}_0) + \mu \mathbf{K}_G(\mathbf{q}_0)] \mathbf{q}_1 = \mathbf{0}, \quad \mathbf{K}_C(\mathbf{q}_0) = \mathbf{K}_{C_0} + \mathbf{K}_C^{NL}(\mathbf{q}_0), \quad (6)$$

gives the multiplier μ of the current stress state, so that

$$\lambda_c = \lambda_b \mu. \quad (7)$$

Convergence is declared when

$$\frac{\lambda_c - \lambda_b}{\lambda_c} \leq \varepsilon_1, \quad (8)$$

with $\varepsilon_1 = 0.005$ by default, and otherwise the load level is advanced by

$$\lambda_b \leftarrow \lambda_b + \eta(\lambda_c - \lambda_b), \quad \eta = 0.8 \text{ on the first step, } \eta = 0.5 \text{ afterwards.} \quad (9)$$

The larger first factor moves quickly into the neighbourhood of the buckling load; the smaller subsequent one avoids overshooting it.

At convergence $\mu \simeq 1$ to within ε_1 , which is what makes the operators of (6) usable directly in the expansion (Section 4).

3.3 Pre-buckling solve near a singular tangent

Each load step solves $\mathbf{f}_{\text{int}}(\mathbf{q}) = \lambda_b \mathbf{f}_0$ by a Newton–Raphson iteration on the tangent stiffness $\mathbf{K}_t = \mathbf{K}_{C_0} + \mathbf{K}_C^{NL}(\mathbf{q}) + \mathbf{K}_G(\mathbf{q})$. Close to the bifurcation point $\mathbf{K}_t$ is nearly singular along the buckling modes, and a load-controlled iteration diverges: any round-off component of the residual in those directions is amplified and throws the iterate off the fundamental path. Four measures are used.

Tangent predictor. The starting point of a step is $\mathbf{q}_0 + \dot{\mathbf{q}}_0 \Delta\lambda$ rather than a rescaling of the previous state.

Deflation of the buckling modes. The fundamental path of a perfect cylinder under axial compression has no component along the buckling modes, which are not axisymmetric. The correction is therefore obtained from the bordered system

$$\begin{bmatrix} \mathbf{K}_t & \mathbf{Q} \\ \mathbf{Q}^\top & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \delta\mathbf{q} \\ \boldsymbol{\alpha} \end{Bmatrix} = \begin{Bmatrix} -\mathbf{R} \\ \mathbf{0} \end{Bmatrix}, \quad (10)$$

with $\mathbf{Q}$ the currently estimated near-critical eigenvectors. This keeps $\delta\mathbf{q}$ on the fundamental path.

Step back-off. If the iteration diverges, η in (9) is halved and the step retried. Divergence is detected early, as soon as the residual measure exceeds 0.5, so that a failed trial costs a few iterations rather than the full iteration budget.

Best iterate on stall. Very close to the bifurcation point the residual of the deflated iteration reaches a floor above the convergence tolerance and then grows again. The best iterate is retained whenever that floor is below a looser acceptance tolerance.

Even with these measures the load-controlled scheme does not always reach $\lambda_b/\lambda_c = 0.995$; the value actually achieved is reported, and the analysis proceeds from the best state obtained. Reaching the target reliably requires arc-length control on the fundamental path, or the extrapolated eigenvalue problem of Sun Eq. (27), neither of which is implemented (Section 9).

3.4 Rates of the pre-buckling state

The first rate follows from Sun Eq. (29), using the tangent stiffness at the converged pre-buckling state,

$$\mathbf{K}_t \dot{\mathbf{q}}_0 = \mathbf{f}_0, \quad (11)$$

solved with the same deflation (10) for the same reason. The second rate follows from Sun Eq. (35) as a backward difference against the load step preceding the converged one,

$$\ddot{\mathbf{q}}_0 = \frac{\dot{\mathbf{q}}_0 - \dot{\mathbf{q}}_0^{\text{prev}}}{\lambda_b - \lambda_b^{\text{prev}}}. \quad (12)$$

Reusing a step the load stepping has already converged on costs nothing, and, unlike a step solved ahead of λ_b , cannot fall on the far side of the bifurcation point. The corresponding stress rates $\dot{\sigma}$ and $\dot{\sigma}$, Sun Eqs. (30) and (34), follow from (3) and (4) through H .

4 Consistency of the operators

The operator appearing in the asymptotic expansion,

$$\phi_2 = \mathbf{K}_C(\mathbf{q}_0) + \mu \mathbf{K}_G(\mathbf{q}_0), \quad (13)$$

must be *the same* operator whose null vector is the buckling mode, that is, exactly the operator of the eigenvalue problem (6). Building ϕ_2 from a differently evaluated $\mathbf{K}_C^{NL}$ than the one used in (6) — for instance from $\mathbf{K}_C^{NL}(\lambda_c \mathbf{q}_0)$ while the eigenvalue problem used $\mathbf{K}_C^{NL}(\mathbf{q}_0)$ — leaves ϕ_2 non-singular along $\mathbf{q}_1$ and corrupts every quantity derived from it. The severity is easy to measure: with the inconsistent pairing,

$$\frac{\|(\mathbf{K}_{C,\text{eig}} + \mu \mathbf{K}_G)\mathbf{q}_1\|}{\|\mathbf{K}_{C_0}\mathbf{q}_1\|} \approx 10^{-10}, \quad \frac{\|\phi_2 \mathbf{q}_1\|}{\|\mathbf{K}_{C_0}\mathbf{q}_1\|} \approx 4.6,$$

i.e. the expansion operator does not annihilate the buckling mode at all. Since $\lambda_b \simeq \lambda_c$ after the iterative algorithm has converged, $\mu \simeq 1$ and (13) is simply the tangent operator at the critical state.

5 Second-order fields and the orthogonality condition

5.1 The singular system

The second-order fields $\mathbf{q}_{ij}$ solve

$$\phi_2 \mathbf{q}_{ij} = \mathbf{g}_{ij}, \quad \mathbf{g}_{ij} = -\frac{1}{2} \phi_{3,ij} - \frac{1}{m} \sum_k a_{kij} \lambda_k \phi_{20,k}, \quad (14)$$

with m the number of Koiter modes. By construction ϕ_2 is singular: its null space is spanned by the buckling modes. Consequently (14) determines $\mathbf{q}_{ij}$ only up to an arbitrary element of that null space, and a plain sparse solve followed by a projection is not a well-posed procedure. Two things are therefore needed: a solver that copes with the singularity, and the correct condition that fixes the null-space component.

5.2 The condition to be imposed

The orthogonality condition of Koiter's theory is Sun Eq. (18),

$$\dot{\sigma}_c \cdot L_{11}(\mathbf{u}_1, \mathbf{u}_2) + \sigma_1 \cdot L_{11}(\dot{\mathbf{u}}_c, \mathbf{u}_2) + \sigma_2 \cdot L_{11}(\dot{\mathbf{u}}_c, \mathbf{u}_1) = 0, \quad (15)$$

whose finite element form is Sun Eq. (33),

$$\mathbf{q}_k^\top [\mathbf{K}_D(\mathbf{q}_b, \dot{\mathbf{q}}_b) + \mathbf{K}_G(\dot{\sigma}_b)] \mathbf{q}_{ij} + \frac{1}{2} \mathbf{q}_k^\top \mathbf{B}_{NL}^\top(\mathbf{q}_k) H \mathbf{B}_{NL}(\mathbf{q}_k) \dot{\mathbf{q}}_b = 0. \quad (16)$$

Two observations turn this into something directly computable.

The linear term is already assembled. The bracket in the first term is the load-parameter derivative of the tangent stiffness,

$$\mathbf{K}_D(\mathbf{q}_b, \dot{\mathbf{q}}_b) + \mathbf{K}_G(\dot{\sigma}_b) = \frac{\partial \phi_2}{\partial \lambda},$$

so the first term is $\mathbf{q}_k^\top (\partial \phi_2 / \partial \lambda) \mathbf{q}_{ij}$, which is exactly the contraction stored as `phi20_a[k]`:

$$\text{first term} = \phi_{20,k} \cdot \mathbf{q}_{ij}. \quad (17)$$

The constant term follows from ϕ_{30} . With $\mathbf{B}_{NL}(\mathbf{q})\mathbf{v} = L_{11}(\mathbf{q}, \mathbf{v})$ the second term of (16) is

$$\frac{1}{2} \langle L_2(\mathbf{u}_k), H L_{11}(\mathbf{u}_k, \dot{\mathbf{u}}_0) \rangle = \frac{1}{2} I, \quad I = \int \langle \mathbf{N}[L_2(\mathbf{u}_k)], L_{11}(\dot{\mathbf{u}}_0, \mathbf{u}_k) \rangle. \quad (18)$$

Now $\phi_{30,ij}$ is assembled as one half of the quadrature weight times the sum of six contractions of the form $\langle \mathbf{N}[\cdot], \partial^2 \boldsymbol{\varepsilon} / \partial \lambda \partial \mathbf{q} \rangle$. When all mode indices coincide those six contributions are equal to one another, by the symmetry of H and of the quadratic form, so that

$$\phi_{30,ij} \cdot \mathbf{q}_k = \frac{1}{2} \cdot 6 I = 3 I \quad \implies \quad \boxed{c_k = \frac{1}{2} I = \frac{\phi_{30,ij} \cdot \mathbf{q}_k}{6}} \quad (19)$$

No new element integral is required. The condition to impose is therefore

$$\phi_{20,k} \cdot \mathbf{q}_{ij} = -c_k. \quad (20)$$

It is weighted by the stiffness matrices, through ϕ_{20} , and inhomogeneous, through c_k . A Gram-Schmidt projection, or equivalently a symmetric Euclidean border $\mathbf{Q}^\top \mathbf{q}_{ij} = \mathbf{0}$, imposes neither of these properties and is the procedure that (20) replaces.

5.3 The bordered system

Both requirements are met by a *non-symmetric* bordered system,

$$\begin{bmatrix} \phi_2 & \mathbf{N}_{sp} \\ \mathbf{W}^\top & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \mathbf{q}_{ij} \\ \boldsymbol{\alpha} \end{Bmatrix} = \begin{Bmatrix} \mathbf{g}_{ij} \\ -\mathbf{c} \end{Bmatrix}, \quad (21)$$

in which the roles of the two borders are distinct:

- the *column* border $\mathbf{N}_{sp}$ spans the null space of ϕ_2 and is what makes the system non-singular;
- the *row* border $\mathbf{W}$ carries the constraints, one column per Koiter mode, $\mathbf{W}[:, k] = \phi_{20,k}$, with the corresponding right-hand side $-c_k$ from (19).

The system is non-singular provided $\mathbf{W}^\top \mathbf{N}_{sp}$ is, and its leading entry $\phi_{20,k} \cdot \mathbf{q}_k$ is exactly the quantity that appears in the denominator of the post-buckling coefficients, hence non-zero by construction.

5.4 Degenerate modes

For a circular cylinder the buckling modes occur in degenerate pairs, one for each sign of the circumferential wave number, so $\dim \ker \phi_2 \geq 2$ even when a single Koiter mode is retained. Two consequences follow.

First, $\mathbf{N}_{sp}$ must span the *whole* near-null space, not only the retained Koiter modes: every eigenvector whose multiplier equals the critical one within a relative tolerance is included, and the set is orthonormalised with a rank-revealing factorisation. Bordering with the Koiter modes alone leaves (21) singular along the remaining directions.

Second, (16) supplies one condition per Koiter mode, which is fewer than $\dim \ker \phi_2$. The degenerate partners lie outside single-mode theory, and the remaining rows of $\mathbf{W}$ keep the Euclidean condition, built from the null-space directions orthogonal to the retained modes. This is a choice, not a result; the consistent treatment of a degenerate critical mode is the multi-mode Koiter expansion.

A related practical point: the basis that an iterative eigensolver returns for a degenerate eigenspace is arbitrary. ARPACK keeps its random seed in a saved variable, so without an

explicit starting vector the basis — and with it $\mathbf{q}_{ij}$ and b_{ijkl} — depends on how many eigenvalue problems have been solved before in the same process. A fixed starting vector is therefore passed to the eigensolver, which makes the results reproducible; it does not remove the underlying ambiguity, which only multi-mode theory resolves.

6 Post-buckling coefficients

With the second-order fields available, the coefficients are

$$a_{ijk} = -\frac{1}{2\lambda_i} \frac{\phi_{3,ij} \cdot \mathbf{q}_k}{\phi_{20,i} \cdot \mathbf{q}_i}, \quad (22)$$

$$b_{ijkl} = -\frac{1}{6\lambda_i(\phi_{20,i} \cdot \mathbf{q}_i)} \left[\phi_{4,ijkl} + 3\phi_{3,ij} \cdot \mathbf{q}_{kl} + 3\phi_{3,il} \cdot \mathbf{q}_{jk} \right. \\ \left. + \lambda_i(a_{iij}\phi_{30,ik} \cdot \mathbf{q}_l + a_{ijk}\phi_{30,il} \cdot \mathbf{q}_i + a_{ikl}\phi_{30,ii} \cdot \mathbf{q}_j) \right. \\ \left. + \phi_{200,ii}\lambda_i^2(a_{iij}a_{ikl} + a_{ijk}a_{ili} + a_{ikl}a_{iij}) \right]. \quad (23)$$

The buckling modes are normalised so that the largest translational amplitude equals the shell thickness, which is the normalisation used by the reference solutions.

7 Algorithm

Algorithm 1 Koiter analysis about a nonlinear pre-buckling state

- 1: solve $\mathbf{K}_{C_0}\mathbf{q} = \mathbf{f}_0$ for the linear pre-buckling state; $\lambda_b \leftarrow 1$
 - 2: **for** $i = 1, \dots, i_{\max}$ **do**
 - 3: assemble $\mathbf{K}_C(\mathbf{q}_0)$, $\mathbf{K}_G(\mathbf{q}_0)$
 - 4: solve the eigenvalue problem (6) $\rightarrow \mu$, $\mathbf{q}_1$; $\lambda_c \leftarrow \lambda_b\mu$
 - 5: **if** (8) holds **then break**
 - 6: $\eta \leftarrow 0.8$ if $i = 1$ else 0.5
 - 7: **repeat**
 - 8: $\lambda_b^{\text{new}} \leftarrow \lambda_b + \eta(\lambda_c - \lambda_b)$
 - 9: predictor $\mathbf{q}_0 + \dot{\mathbf{q}}_0(\lambda_b^{\text{new}} - \lambda_b)$; Newton–Raphson with deflation (10)
 - 10: **on failure** $\eta \leftarrow \eta/2$
 - 11: **until** converged or η exhausted
 - 12: store $(\lambda_b, \dot{\mathbf{q}}_0)$ as the previous step; advance
 - 13: **end for**
 - 14: $\dot{\mathbf{q}}_0$ from (11); $\ddot{\mathbf{q}}_0$ from (12)
 - 15: assemble $\phi_2, \phi_3, \phi_4, \phi_{20}, \phi_{30}, \phi_{200}$ using (3)–(5)
 - 16: a_{ijk} from (22)
 - 17: $\mathbf{q}_{ij}$ from the bordered system (21) with the weighted condition (20)
 - 18: b_{ijkl} from (23)
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8 Effect on the results

The two verification cases are the composite cylinder of Sun §3.1 and the NASA shell AW-CYL-1-1 of Arbocz, Starnes and Nemeth (AIAA-2001-1392); they are the same shell, so the ANILISA n -search of the latter provides reference values for both. Buckling loads are normalised by $N_{cl} = E_{11}t^2/(R\sqrt{3(1-\nu_{12}^2)})$.

	expanding at N_{xx}^{unit}	about $\lambda_b \simeq \lambda_c$	ANILISA, nonlinear pre-buckling
λ_b/λ_c	0.14	0.985	—
N_{cr}/N_{cl}	0.3581	0.3367	0.337088 ($n = 7$)
$\max w_0 /t$	0.026	0.192	—

The first column is the membrane pre-buckling result, which is what the computation returns when the expansion point is left at the reference load; the reference value for that regime is 0.365992. The second column agrees with the ANILISA nonlinear pre-buckling entry for the same circumferential wave number to 0.1%.

The post-buckling coefficient is far more sensitive to mesh refinement than the buckling load, and the reference value $b = -0.376$ belongs to the $n = 11$ edge-buckling mode, so a like-for-like comparison requires a mesh fine enough for that mode to become critical. On the finest mesh run here the magnitude $|b| = 0.437$ is of the right order, against -0.37605 from ANILISA and -0.3772 from Sun Table 2, where expanding at the reference load had given -0.0598 , close to the membrane pre-buckling value -0.048844 .

Its *sign*, however, is not determined. Two runs of the same mesh that differ only in the member of the degenerate pair returned by the eigensolver gave -0.43100 and $+0.43731$, with λ_c , the wave number and λ_b/λ_c identical. Since $b < 0$ means an imperfection-sensitive shell and $b > 0$ the opposite, this is a substantive limitation of a single-mode expansion about a degenerate critical mode, not solver noise; see Section 9.

9 Known limitations

- 1. Load control near the bifurcation point.** The scheme of Section 3.3 reaches λ_b/λ_c between about 0.93 and 0.985 rather than the 0.995 that Sun §2.2 asks for. Arc-length control on the fundamental path, or the extrapolated eigenvalue problem of Sun Eq. (27), $[\mathbf{K}_t + (\lambda_c - \lambda_b) \partial \mathbf{K}_t / \partial \lambda] \mathbf{q}_1 = 0$, would remove the need to sit on top of the bifurcation point.
- 2. Degenerate critical modes.** As discussed in Section 5.4, the condition (20) supplies one equation per retained mode while the null space is larger. The observable consequence is that the sign of b_{ijkl} depends on which member of the degenerate pair is retained: the same mesh yields -0.43100 or $+0.43731$ according to the basis the eigensolver returns. Fixing the eigensolver starting vector makes a given run reproducible but does not resolve the ambiguity. Imposing (20) in place of the Euclidean condition is the first thing to try, since it is what determines the null-space component of $\mathbf{q}_{ij}$; multi-mode Koiter is the consistent treatment.
- 3. Symmetrisation of ϕ_{30} for distinct modes.** The identity (19) was established for coinciding mode indices, which covers the single-mode case exercised here; the general multi-mode symmetrisation remains to be worked out.
- 4. Quadrature.** Convergence with respect to the number of Gauss points has been verified: $n_{int} = 4$ gives b to within 0.4% of $n_{int} = 5$ and 6, while $n_{int} = 2$ under-integrates and produces a rank-deficient element.